

System Overview: An online marketplace enabling independent craftspeople to set up storefronts, manage product catalogs, receive orders, and receive automated split payouts with platform commission deductions. • **Target Actors:** Shopper, Artisan Vendor, Payment Gateway (External System)

REQ ID	TYPE	DESCRIPTION	PRIORITY	ACCEPTANCE CRITERIA	RATIONALE	COMMENTS
FUNCTIONAL REQUIREMENTS (FR-001 TO FR-005)						
FR-001	Functional	The system shall split customer cart payments at checkout, allocating respective item earnings to multiple independent vendor accounts after deducting a 5% platform fee.	High	Pass: Split payout calculations balance to total cart value. Fail: Payout calculation discrepancies.	Ensures accurate, automated financial distribution to independent artisans while securing platform transaction revenue (Given in PS #31).	Core monetization requirement; calculations must balance to 100% of order total.
FR-002	Functional	The system shall allow artisan vendors to set up and manage an independent storefront profile, including artisan bio, shop policies, and linked payout disbursement account details.	High	Pass: Storefront profile updates and verified payout details are successfully saved and rendered publicly on the artisan's storefront page. Fail: Missing mandatory profile fields or unvalidated payout details prevent storefront publishing.	Enables independent craftspeople to establish their distinct brand identity and receive automated financial payouts.	Peer: Validated banking credentials required before activating vendor storefront.
FR-003	Functional	The system shall allow artisan vendors to create, update, and manage handcrafted product listings with titles, descriptions, pricing, inventory quantities, and high-resolution product media.	High	Pass: New or updated product listings appear in the artisan's catalog and marketplace search within 5 seconds with accurate stock counts. Fail: Listings with invalid pricing (≤ 0) or missing mandatory attributes fail validation and are rejected.	Empowers artisans to maintain an up-to-date catalog of handcrafted items and prevent overselling through inventory synchronization.	Real-time stock validation prevents race conditions during simultaneous customer checkouts.
FR-004	Functional	The system shall allow shoppers to add handcrafted items from multiple independent artisan vendors into a unified cart and execute a single consolidated checkout order.	High	Pass: Cart aggregates items across distinct vendors, calculates itemized totals, applies applicable coupon discounts, and initiates payment authorization. Fail: Cart fails to itemize multi-vendor items or calculation mismatch occurs between item sums and total payable amount.	Provides a seamless purchasing experience for shoppers buying from multiple artisans simultaneously without requiring separate checkout transactions.	Consolidated order is decomposed into vendor sub-orders upon payment authorization.
FR-005	Functional	The system shall notify artisan vendors of confirmed customer orders containing their respective products and allow vendors to update fulfillment status (Processing, Dispatched, Delivered) with carrier tracking details.	Medium	Pass: Vendor dashboard immediately displays incoming sub-orders upon successful checkout and status transitions trigger automated notifications to the shopper. Fail: Order details fail to isolate vendor-specific items or status updates fail to persist.	Enables independent artisans to fulfill customer orders independently while keeping shoppers informed of dispatch status.	Status changes send transactional push/email notifications with courier tracking links.
NON-FUNCTIONAL REQUIREMENTS (NFR-001 & NFR-002)						
NFR-001	Performance & Security	The product catalog must support high-resolution image rendering with CDN caching delivering load times < 500 ms.	High	Pass: Benchmarking tests confirm target latency (< 500 ms) and security standards under simulated peak load. Fail: Catalog page load latency ≥ 500 ms or CDN caching failure under peak load.	Ensures fast page loading for image-heavy handcrafted goods, preserving user engagement and minimizing bounce rates (Given in PS #31).	Edge CDN nodes cache optimized image formats (WebP/AVIF); benchmarked via Lighthouse.
NFR-002	Security & Compliance	The system shall encrypt all sensitive financial transactions and payout account data in transit using TLS 1.3 and at rest using AES-256 encryption, adhering to PCI-DSS Level 1 compliance standards.	High	Pass: Automated vulnerability scans and penetration audits confirm 100% encryption coverage for payment payloads with zero plain-text storage of payment credentials. Fail: Any unencrypted transmission or unmasked storage of sensitive banking/card data detected.	Protects shoppers' financial credentials and vendors' banking payout information against unauthorized access, data breaches, and regulatory penalties.	Immutable append-only ledger verifies double-entry reconciliation between payouts and fees.